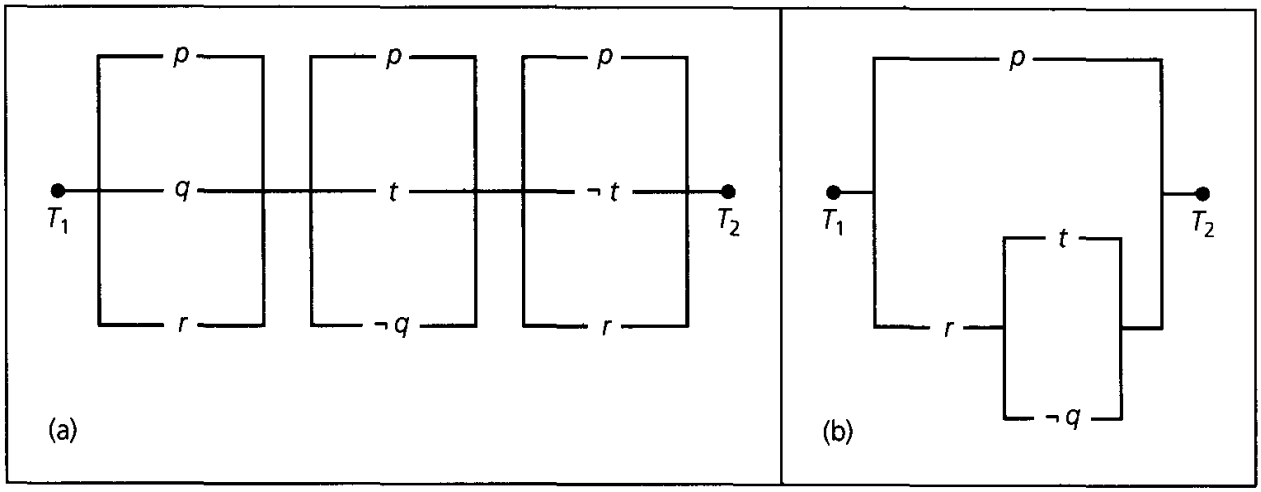



**Figure 2.3**

Hence $(p \vee q \vee r) \wedge (p \vee t \vee \neg q) \wedge (p \vee \neg t \vee r) \Leftrightarrow p \vee [r \wedge (t \vee \neg q)]$, and the network shown in Fig. 2.3(b) is equivalent to the original network in the sense that current flows from $T_1$ to $T_2$ in network (a) exactly when it does so in network (b). But in (b) the network has only four switches, five fewer than in network (a).

## EXERCISES 2.2

1. Let $p$, $q$, $r$ denote primitive statements.
   a) Use truth tables to verify the following logical equivalences.
      i) $p \rightarrow (q \wedge r) \Leftrightarrow (p \rightarrow q) \wedge (p \rightarrow r)$
      ii) $[(p \vee q) \rightarrow r] \Leftrightarrow [(p \rightarrow r) \wedge (q \rightarrow r)]$
      iii) $[p \rightarrow (q \vee r)] \Leftrightarrow [\neg r \rightarrow (p \rightarrow q)]$
   b) Use the substitution rules to show that $[p \rightarrow (q \vee r)] \Leftrightarrow [(p \wedge \neg q) \rightarrow r]$.
2. Verify the first Absorption Law by means of a truth table.
3. Use the substitution rules to verify that each of the following is a tautology. (Here $p$, $q$, and $r$ are primitive statements.)
   a) $[p \vee (q \wedge r)] \vee \neg [p \vee (q \wedge r)]$
   b) $[(p \vee q) \rightarrow r] \leftrightarrow [\neg r \rightarrow \neg (p \vee q)]$
   c) $[[(p \vee q) \rightarrow r] \vee (s \wedge t)] \leftrightarrow [[[(p \vee q) \rightarrow r] \vee s] \wedge [[(p \vee q) \rightarrow r] \vee t]]$
4. For primitive statements $p$, $q$, $r$, and $s$, simplify the compound statement
$$[[[(p \wedge q) \wedge r] \vee [(p \wedge r) \wedge \neg r]] \vee \neg q] \rightarrow s.$$
5. Negate and express each of the following statements in smooth English.
   a) Kelsey will get a good education if she puts her studies before her interest in cheerleading.
   b) Norma is doing her mathematics homework, and Karen is practicing her piano lessons.
   c) If Logan goes on vacation, then he will enjoy himself if he doesn't worry about traveling by airplane.
   d) If Harold passes his Pascal course and finishes his data structures project, then he will graduate at the end of the semester.

**6.** Negate each of the following, and simplify the resulting statement.
**a)** $p \wedge (q \vee r) \wedge (\neg p \vee \neg q \vee r)$ **b)** $(p \wedge q) \rightarrow r$
**c)** $p \rightarrow (\neg q \wedge r)$ **d)** $p \vee q \vee (\neg p \wedge \neg q \wedge r)$

**7.** **a)** If $p$, $q$ are primitive statements, prove that $(\neg p \vee q) \wedge (p \wedge (p \wedge q)) \Leftrightarrow (p \wedge q)$.
**b)** Write the dual of the logical equivalence in part (a).

**8.** Write the dual for (a) $q \rightarrow p$, (b) $p \rightarrow (q \wedge r)$, (c) $p \leftrightarrow q$, and (d) $p \veebar q$, where $p$, $q$, and $r$ are primitive statements.

**9.** Write the converse, inverse, and contrapositive of each of the following implications. For each implication, determine its truth value as well as the truth values of its corresponding converse, inverse, and contrapositive.
**a)** If today is Labor Day, then tomorrow is Tuesday.
**b)** If $-1 < 3$ and $3 + 7 = 10$, then $\sin\left(\frac{3\pi}{2}\right) = -1$.
**c)** If Wesley lives in New England, then Wesley lives in Vermont.

**10.** Determine whether each of the following is true or false. Here $p$, $q$, and $r$ are arbitrary statements.
**a)** An equivalent way to express the converse of "$p$ is sufficient for $q$" is "$p$ is necessary for $q$."
**b)** An equivalent way to express the inverse of "$p$ is necessary for $q$" is "$\neg q$ is sufficient for $\neg p$."
**c)** An equivalent way to express the contrapositive of "$p$ is necessary for $q$" is "$\neg q$ is necessary for $\neg p$."
**d)** An equivalent way to express the converse of $p \rightarrow (q \rightarrow r)$ is $(\neg q \vee r) \rightarrow p$.

**11.** Let $p$, $q$, and $r$ denote primitive statements. Find a form of the contrapositive of $p \rightarrow (q \rightarrow r)$ with (a) only one occurrence of the connective $\rightarrow$; (b) no occurrences of the connective $\rightarrow$.

**12.** Write the converse, inverse, contrapositive, and negation of the following statement: "If Sandra finishes her work, she will go to the basketball game unless it snows."

**13.** Consider the two Pascal program segments shown in Fig. 2.1 (Example 2.16). Find the total number of comparisons executed in each segment if, instead of being assigned the value 4, $z$ were assigned the value (a) 2; (b) 6; (c) 9; (d) 10; (e) 15; (f) $n$, an integer greater than 10.

**14.** Show that for primitive statements $p$, $q$,

$$p \veebar q \Leftrightarrow [(p \wedge \neg q) \vee (\neg p \wedge q)] \Leftrightarrow \neg(p \leftrightarrow q).$$

**15.** Verify that $[(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)] \Leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)]$, for primitive statements $p$, $q$, and $r$.

**16.** For primitive statements $p$, $q$,
**a)** verify that $p \rightarrow [q \rightarrow (p \wedge q)]$ is a tautology.
**b)** verify that $(p \vee q) \rightarrow [q \rightarrow q]$ is a tautology by using the result from part (a) along with the substitution rules and the laws of logic.
**c)** is $(p \vee q) \rightarrow [q \rightarrow (p \wedge q)]$ a tautology?

**17.** Define the connective "Nand" or "Not... and..." by $(p \uparrow q) \Leftrightarrow \neg(p \wedge q)$, for any statements $p$, $q$. Represent the following using only this connective.
**a)** $\neg p$ **b)** $p \vee q$ **c)** $p \wedge q$
**d)** $p \rightarrow q$ **e)** $p \leftrightarrow q$

**18.** The connective "Nor" or "Not... or..." is defined for any statements $p$, $q$ by $(p \downarrow q) \Leftrightarrow \neg(p \vee q)$. Represent the statements in parts (a) through (e) of Exercise 17, using only this connective.

**19.** For any statements $p$, $q$, prove that

**a)** $\neg(p \downarrow q) \Leftrightarrow (\neg p \uparrow \neg q)$ **b)** $\neg(p \uparrow q) \Leftrightarrow (\neg p \downarrow \neg q)$

**20.** Give the reasons for each step in the following simplifications of compound statements.

**a)**

| | Reasons |
|---|---|
| $[(p \vee q) \wedge (p \vee \neg q)] \vee q$ | |
| $\Leftrightarrow [p \vee (q \wedge \neg q)] \vee q$ | |
| $\Leftrightarrow (p \vee F_0) \vee q$ | |
| $\Leftrightarrow p \vee q$ | |

**b)**

| | Reasons |
|---|---|
| $\neg(p \vee q) \vee [(\neg p \wedge q) \vee \neg q]$ | |
| $\Leftrightarrow \neg(p \vee q) \vee [\neg q \vee (\neg p \wedge q)]$ | |
| $\Leftrightarrow \neg(p \vee q) \vee [(\neg q \vee \neg p) \wedge (\neg q \vee q)]$ | |
| $\Leftrightarrow \neg(p \vee q) \vee [(\neg q \vee \neg p) \wedge T_0]$ | |
| $\Leftrightarrow \neg(p \vee q) \vee (\neg q \vee \neg p)$ | |
| $\Leftrightarrow \neg(p \vee q) \vee \neg(q \wedge p)$ | |
| $\Leftrightarrow \neg[(p \vee q) \wedge (q \wedge p)]$ | |
| $\Leftrightarrow \neg[(q \wedge p) \wedge (p \vee q)]$ | |
| $\Leftrightarrow \neg[q \wedge [p \wedge (p \vee q)]]$ | |
| $\Leftrightarrow \neg(q \wedge p)$ | |

**c)**

| | Reasons |
|---|---|
| $(p \rightarrow q) \wedge [\neg q \wedge (r \vee \neg q)]$ | |
| $\Leftrightarrow (p \rightarrow q) \wedge \neg q$ | |
| $\Leftrightarrow (\neg p \vee q) \wedge \neg q$ | |
| $\Leftrightarrow \neg q \wedge (\neg p \vee q)$ | |
| $\Leftrightarrow (\neg q \wedge \neg p) \vee (\neg q \wedge q)$ | |
| $\Leftrightarrow (\neg q \wedge \neg p) \vee F_0$ | |
| $\Leftrightarrow \neg q \wedge \neg p$ | |
| $\Leftrightarrow \neg(q \vee p)$ | |

**21.** Provide the steps and reasons, as in Exercise 20, to establish the following logical equivalences.

**a)** $p \vee [p \wedge (p \vee q)] \Leftrightarrow p$

**b)** $p \vee q \vee (\neg p \wedge \neg q \wedge r) \Leftrightarrow p \vee q \vee r$

**c)** $[(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r)] \Leftrightarrow p \wedge q$

**d)** $p \wedge [(\neg q \rightarrow (r \wedge r)) \vee \neg[q \vee ((r \wedge s) \vee (r \wedge \neg s))]] \Leftrightarrow p$

**22.** Simplify each of the networks shown in Fig. 2.4.

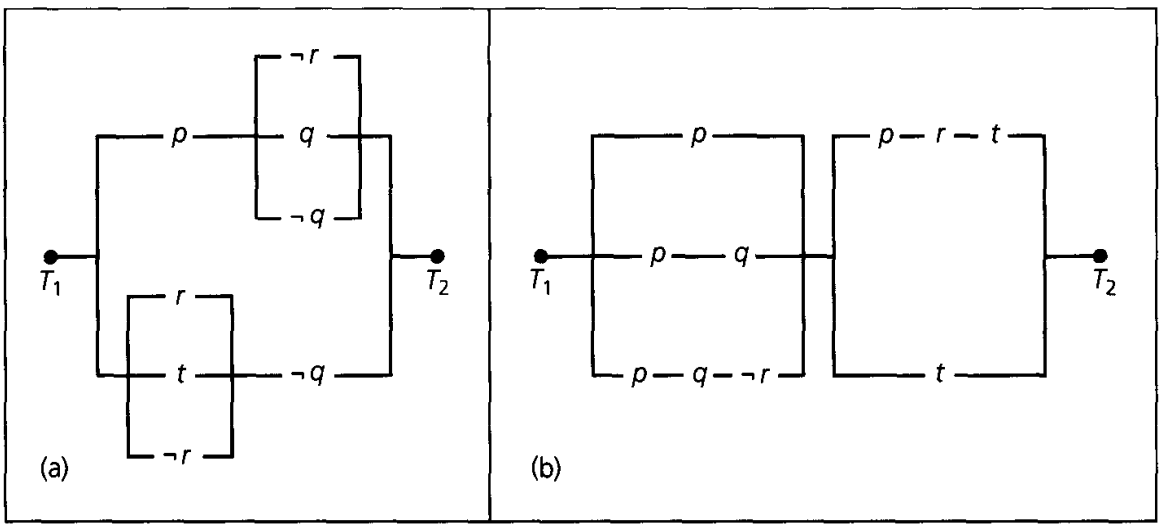



**Figure 2.4**